

LECTURE 10: TESTING IN LARGE SAMPLES

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REFERENCE CB, Chapter 10.3, or HMC, Chapter 6.3, 6.5

We observe a random sample $X = (X_1, \dots, X_n)$ from a distribution with density $f(x | \theta)$, $\theta \in \mathbb{R}$. Let $\mathcal{L}_n(\theta | X) = \prod_{i=1}^n f(X_i | \theta)$ denote the likelihood function. Consider testing the null $H_0: \theta = \theta_0$ against the two-sided alternative $H_1: \theta \neq \theta_0$. We develop three tests of these hypotheses that are justified by large-sample arguments: the likelihood ratio (LR) test, the Lagrange multiplier (LM) test, and the Wald test. It will turn out that while these tests are different in finite samples, they are asymptotically equivalent. We then generalize to the case where θ is a vector, and discuss the asymptotic optimality of this trinity of tests.

1. MAXIMIZED LIKELIHOOD RATIO TEST

We have seen that when the alternative is simple, $H_1: \theta = \theta_1$, then rejecting for large values of the LR, $\mathcal{L}_n(\theta_1 | X) / \mathcal{L}_n(\theta_0 | X)$, is a good idea—in fact, the best possible idea. The alternative $H_1: \theta \neq \theta_0$ is not simple, but the idea that we should compare the likelihood under the null with that under the alternative still seems appealing. Since there are many likelihood values under the alternative, we pick the value that makes the alternative look the most favorable; that is, we pick $\sup_{\theta \in \Theta} \mathcal{L}_n(\theta | X) = \mathcal{L}_n(\hat{\theta}_{ML} | X)$, where $\hat{\theta}_{ML}$ is the maximum likelihood estimator.¹ This leads to the maximized LR statistic

$$\text{MLR}(X) = \frac{\mathcal{L}_n(\hat{\theta}_{ML} | X)}{\mathcal{L}_n(\theta_0 | X)}.$$

Rejecting for large values of $\text{MLR}(X)$ is equivalent to rejecting for large values of

$$\zeta_{LR} = 2 \log \text{MLR}(X) = 2(\ell_n(\hat{\theta}_{ML}) - \ell_n(\theta_0)), \quad (1)$$

where $\ell_n(\theta)$ is the log-likelihood. Typically, but confusingly, people omit “maximized” and refer to this test as the “likelihood ratio” test. However, unlike the LR test (the

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1. Since θ_0 is single point, provided the likelihood is continuous, $\sup_{\theta \in \Theta} \mathcal{L}_n(\theta | X) = \sup_{\theta \in \Theta \setminus \{\theta_0\}} \mathcal{L}_n(\theta | X)$.

use of which is justified by the Neyman-Pearson lemma), the maximized LR test has no finite-sample justification. The test is due to Neyman and Pearson (1928).

The next result allows us to compute the appropriate critical value that will ensure that the test satisfies the level constraint in large samples.

Theorem 1 (HMC, Theorem 6.3.1, CB, Theorem 10.3.1). Under the null hypothesis and regularity conditions similar to those leading to normality of maximum likelihood estimator (MLE), $\zeta_{LR} \Rightarrow \chi^2(1)$.

Proof. One of the regularity conditions is that θ_0 lies in the interior of the parameter space Θ . Since $\hat{\theta}_{ML}$ is consistent, the estimator must also lie in the interior of the parameter space with probability approaching one. In large samples, it must therefore hold that $\ell'_n(\hat{\theta}_{ML}) = 0$. Therefore, if we approximate $\ell_n(\theta_0)$ using an expansion around $\hat{\theta}_{ML}$, we obtain, for some $\tilde{\theta}$ between $\hat{\theta}_{ML}$ and θ_0 ,

$$\ell_n(\theta_0) = \ell_n(\hat{\theta}_{ML}) + \frac{1}{2} \ell''_n(\tilde{\theta})(\theta_0 - \hat{\theta}_{ML})^2. \quad (2)$$

Since $\hat{\theta}_{ML} \xrightarrow{P} \theta_0$ by MLE theory, we have $\tilde{\theta} \xrightarrow{P} \theta_0$. Furthermore, $\ell''_n(\theta_0)/n \xrightarrow{P} -\mathcal{I}(\theta_0)$ by law of large numbers (LLN). This suggests, but does not imply, that

$$\ell''_n(\tilde{\theta})/n \xrightarrow{P} -\mathcal{I}(\theta_0). \quad (3)$$

As with the proof of asymptotic normality of $\hat{\theta}_{ML}$, the problem is that we have a sequence of functions $\frac{1}{n} \ell''_n(\theta)$ instead of just one function, so we cannot just apply the continuous mapping theorem (CMT). Establishing eq. (3) requires the concept of asymptotic equicontinuity, which we do not cover in this class. Suppose we nonetheless believe this result. We may then rewrite eq. (2) as

$$\zeta_{LR} = 2(\ell_n(\hat{\theta}_{ML}) - \ell_n(\theta_0)) = -\ell''_n(\tilde{\theta}) \cdot (\theta_0 - \hat{\theta}_{ML})^2 = -\frac{\ell''_n(\tilde{\theta})}{n\mathcal{I}(\theta_0)} \cdot \mathcal{I}(\theta_0)n(\theta_0 - \hat{\theta}_{ML})^2.$$

By eq. (3), the first term converges in probability to 1, and by the CMT and MLE normality, the second term converges in distribution to $\chi^2(1)$. The conclusion thus follows by Slutsky's theorem. $\square$

Since the $1 - \alpha$ quantile of a $\chi^2(1)$ distribution is given by $z_{1-\alpha/2}$, the $1 - \alpha/2$ quantile of a standard normal distribution, it follows from the theorem that the test that rejects when $\zeta_{LR} \geq z_{1-\alpha/2}^2$ has size α in large samples, in the sense that the null rejection probability converges to α . In finite samples, the size of this test may be greater or smaller than α .

Example 1. Let $X_1, \dots, X_n$ be a random sample from a $\mathcal{N}(\theta, \sigma^2)$ distribution, with σ^2 known. Recall that $\hat{\theta}_{ML} = \bar{X}_n$, the sample mean. The maximized LR test for testing the null $H_0: \theta = \theta_0$ against $H_1: \theta \neq \theta_0$ is thus given by

$$\begin{aligned} \zeta_{LR} &= 2(\ell_n(\bar{X}_n) - \ell_n(\theta_0)) \\ &= -\frac{1}{\sigma^2} \sum_i (X_i - \bar{X}_n)^2 + \frac{1}{\sigma^2} \sum_i (X_i - \theta_0)^2 \\ &= \frac{n}{\sigma^2} (\bar{X}_n^2 + \theta_0^2 - 2\theta_0 \bar{X}_n) = \frac{n}{\sigma^2} (\bar{X}_n - \theta_0)^2. \end{aligned}$$

The LR test rejects if $\frac{n}{\sigma^2} (\bar{X}_n - \theta_0)^2 \geq z_{1-\alpha/2}^2$. Note that this is actually the uniformly most powerful (UMP) unbiased test we derived in Lecture 9, which controls size in finite

samples. □

Example 2. Let $X_1, \dots, X_n$ be a random sample from a $\text{Poisson}(\theta)$ distribution, which has density $\theta^x e^{-\theta} / x!$, for $x = 0, 1, \dots$ and $\theta > 0$. The log-likelihood is

$$\ell_n(\theta) = n\bar{X}_n \log(\theta) - n\theta - \sum_{i=1}^n \log(X_i!),$$

which implies that $\hat{\theta}_{ML} = \bar{X}_n$, and

$$\begin{aligned} \zeta_{LR} &= 2(n\bar{X}_n \log(\hat{\theta}_{ML}/\theta_0) - n\hat{\theta}_{ML} + n\theta_0) \\ &= 2n(\bar{X}_n \log(\bar{X}_n/\theta_0) - (\bar{X}_n - \theta_0)) = 2(S \log(S/\tilde{\theta}_0) - S + \tilde{\theta}_0), \end{aligned}$$

where θ_0 is the null value, $S = \sum_{i=1}^n X_i$, and $\tilde{\theta}_0 = n\theta_0$. Observe that the test statistic $\zeta_{LR} = \zeta_{LR}(S)$ is strictly convex in S , with minimum at $S = \tilde{\theta}_0$. As a result, the acceptance region $\{S \geq 0: \zeta_{LR}(S) \leq z_{1-\alpha/2}^2\}$ will be an interval around $\tilde{\theta}_0$. Because $\lim_{x \rightarrow 0} x \log(x) = 0$, we have $\zeta_{LR}(0) = 2\tilde{\theta}_0$. Hence, if $\tilde{\theta}_0 \leq z_{1-\alpha/2}^2/2$, $S = 0$ is accepted, and the interval has the form $[0, U(\tilde{\theta}_0)]$. Otherwise, the acceptance region has the form $[L(\tilde{\theta}_0), U(\tilde{\theta}_0)]$ with $L(\tilde{\theta}_0) > 0$. The functions U and L can be solved for numerically.

Since $S \sim \text{Poisson}(\tilde{\theta}_0)$ under the null, we can numerically compute the exact finite-sample size of the test by computing $1 - P(S \in [L(\tilde{\theta}_0), U(\tilde{\theta}_0)])$. Note that the size depends on n and θ_0 only through their product, $\tilde{\theta}_0 = n\theta_0$. Figure 1 plots the test size. Can you explain the spike at $\tilde{\theta}_0 = 1.921$ and the erratic behavior of the test size?

As a numerical example, suppose $\theta_0 = 6$, $\bar{X}_n = 5$, and $n = 100$. Then

$$\zeta_{LR} = 200(5 \log(5/6) + 1) \approx 17.7.$$

The critical value is 1.96^2 . So the test rejects. □

2. WALD TEST

In the proof of Theorem 1, we have shown that the LR statistic equals

$$\zeta_{LR} = \left(-\frac{\ell_n''(\tilde{\theta})}{n} \right) n(\theta_0 - \hat{\theta}_{ML})^2, \quad (4)$$

where $\tilde{\theta}$ is between θ_0 and $\hat{\theta}_{ML}$. We can think of $-\ell_n''(\tilde{\theta})/n$ as an estimate of the information matrix. Alternatively, we can estimate the information matrix by evaluating ℓ_n'' at $\hat{\theta}_{ML}$, which yields the Wald (1943) statistic

$$\zeta_W = \left(-\frac{\ell_n''(\hat{\theta}_{ML})}{n} \right) n(\theta_0 - \hat{\theta}_{ML})^2. \quad (5)$$

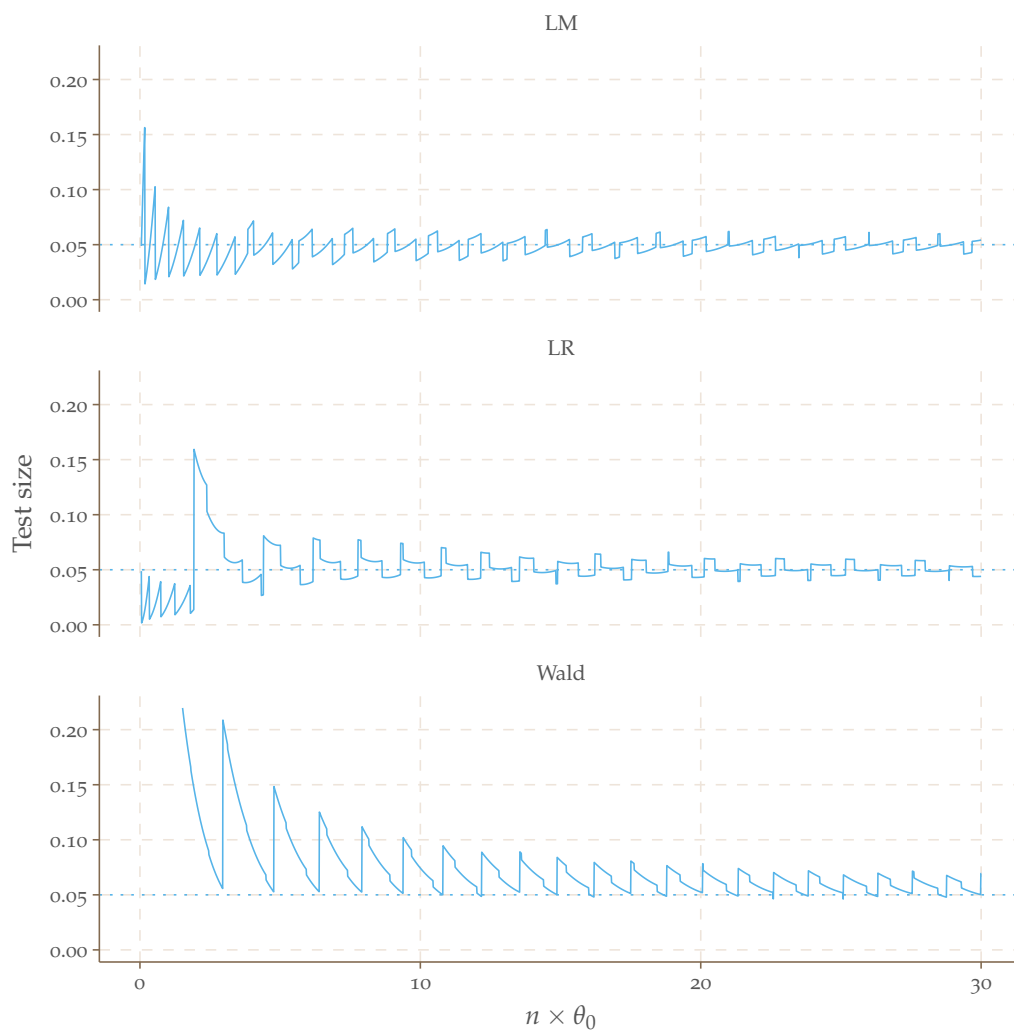


Figure 1: Exact size of the LM, LR, and Wald tests for the Poisson distribution (Example 2).

This has the advantage that it doesn't involve any computation under the null. Since both $\hat{\theta}_{ML}$ and $\tilde{\theta}$ converge in probability to θ_0 under the null, this substitution has negligible effect in large samples. As a result,

$$\zeta_W - \zeta_{LR} \xrightarrow{p_{H_0}} 0,$$

so that the $z_{1-\alpha/2}^2$ critical value is again appropriate, and the LR and Wald statistics are asymptotically equivalent. However, the tests differ in finite samples, unless $\ell_n''(\theta)$ doesn't depend on θ (when is that the case?).

Remark 1. The Wald statistic, unlike the LR statistic, is not invariant to reparametrization.

Another way of motivating this statistic is to note that, under the null

$$\sqrt{n}(\hat{\theta}_{ML} - \theta_0) \Rightarrow \mathcal{N}(0, \mathcal{I}(\theta_0)^{-1}),$$

and that $(-\frac{\ell_n''(\hat{\theta}_{ML})}{n})^{-1}$ is a consistent estimator of $\mathcal{I}(\theta_0)^{-1}$. More generally, whenever we have an estimator $\hat{\theta}$ such that $\sqrt{n}(\hat{\theta} - \theta) \Rightarrow \mathcal{N}(0, V)$, and a consistent estimator $\hat{V}$ of the asymptotic variance V , then by Slutsky's theorem, under the null $\theta = \theta_0$,

$$\sqrt{n}(\hat{\theta} - \theta_0) / \sqrt{\hat{V}} \Rightarrow \mathcal{N}(0, 1),$$

so that an asymptotically level α test rejects the null if

$$\frac{n(\hat{\theta} - \theta_0)^2}{\hat{V}} \geq z_{1-\alpha/2}^2.$$

If the alternative hypothesis is one-sided, $\Theta_1 = \{\theta : \theta > \theta_0\}$, then it is better (more powerful) to reject whenever

$$\frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\sqrt{\hat{V}}} \geq z_{1-\alpha}.$$

Example 1 (continued). Here $\zeta_W = \zeta_{LR}$. ☒

Example 2 (continued). Differentiating the log-likelihood yields

$$\ell_n'(\theta) = n\bar{X}_n/\theta - n, \quad \ell_n''(\theta) = -n\bar{X}_n/\theta^2.$$

Thus,

$$\mathcal{I}(\theta) = E \left[-\frac{1}{n} \ell_n''(\theta) \right] = \frac{1}{\theta},$$

and the Wald statistic uses the estimator $-\frac{1}{n} \ell_n''(\hat{\theta}_{ML}) = 1/\hat{\theta}_{ML}$, so that

$$\zeta_W = \frac{n(\hat{\theta}_{ML} - \theta_0)^2}{\hat{\theta}_{ML}} = \frac{(S - \tilde{\theta}_0)^2}{S}.$$

The test thus rejects if $(S - \tilde{\theta}_0)^2 > z_{1-\alpha/2}^2 S$, which is a quadratic equation in S . Solving this quadratic equation yields that the acceptance region is given by $\tilde{\theta}_0 + z_{1-\alpha/2}^2/2 \pm$

$z_{1-\alpha/2}\sqrt{\tilde{\theta}_0 + z_{1-\alpha/2}^2/4}$. Figure 1 plots the test size. The test is noticeably size-distorted, particularly for small $\tilde{\theta}_0$. The issue is that when $S = 0$, the Wald test statistic is infinite for any $\tilde{\theta}_0$, so the Wald test rejects. In our numerical example,

$$\zeta_W = 100 \cdot (5 - 6)^2 \cdot (1/5) = 20.$$

So the test based on the Wald statistic rejects the null hypothesis with an even smaller p -value than the test based on the LR statistic. $\square$

3. SCORE TEST

Recall that the score is defined as the derivative of the log-likelihood,

$$\mathcal{S}_n(\theta) = \ell'_n(\theta) = \sum_{i=1}^n \frac{\partial \log f(X_i | \theta)}{\partial \theta},$$

and that by the first information equality and the central limit theorem (CLT), the score is asymptotically normal $n^{-1/2}\mathcal{S}_n(\theta) \Rightarrow \mathcal{N}(0, \mathcal{I}(\theta))$. Hence, by the CMT, $\mathcal{S}_n(\theta)^2/n\mathcal{I}(\theta) \Rightarrow \chi^2(1)$.

By the law of large numbers, $-\ell''_n(\theta)/n \xrightarrow{p} \mathcal{I}(\theta)$. Therefore, by Slutsky's lemma, under the null hypothesis

$$\zeta_{LM} = \frac{\mathcal{S}_n(\theta_0)^2}{-\ell''_n(\theta_0)} \Rightarrow \chi^2(1). \quad (6)$$

The test that rejects whenever $\zeta_{LM} > z_{1-\alpha/2}^2$ is known as Rao's (1948) score test, or the Lagrange multiplier (LM) test. The latter name is due to Silvey (1959). One can also use the Fisher information instead of the Hessian, rejecting whenever $\tilde{\zeta}_{LM} = \mathcal{S}_n(\theta_0)^2 / -\ell''_n(\theta_0) > z_{1-\alpha/2}^2$. The two versions are asymptotically equivalent.

To see why Silvey called it the LM test, set up the Lagrangian for maximizing the likelihood subject to the null constraint $\theta = \theta_0$,

$$\ell_n(\theta) - \lambda(\theta - \theta_0).$$

(Here the maximization is trivial, but the idea of the LM test extends to cases in which the null is composite, as we discuss below). The first-order condition is

$$\mathcal{S}_n(\theta_0) = \lambda.$$

We know that $\hat{\theta}_{ML}$ maximizes $\ell_n(\theta)$ and, in large samples, $\hat{\theta}_{ML}$ will be close to the true parameter value θ with large probability. If the true value of the parameter θ is far from θ_0 , then $\hat{\theta}_{ML}$ will be far from θ_0 . Thus, in order for θ_0 to satisfy the Lagrangian's first-order condition, λ would need to be large. As a result, $\mathcal{S}_n(\theta_0)$ and ζ_{LM} will both be large.

In fact, $\zeta_{LM} - \zeta_{LR} \xrightarrow{p} 0$, so that this statistic is asymptotically equivalent to the Wald and LR statistics.

Proof. To show this result, note that by the mean value theorem,

$$\mathcal{S}_n(\theta_0) = \mathcal{S}_n(\theta_0) - \mathcal{S}_n(\hat{\theta}_{ML}) = \ell_n''(\bar{\theta})(\theta_0 - \hat{\theta}_{ML}),$$

where $\bar{\theta}$ is between θ_0 and $\hat{\theta}_{ML}$. Thus, $\zeta_{LM} = -(\theta_0 - \hat{\theta}_{ML})^2 \ell_n''(\bar{\theta})^2 / \ell_n''(\theta_0)$. Therefore, using eq. (4), we have

$$\zeta_{LM} - \zeta_{LR} = \left(\frac{(\ell_n''(\bar{\theta})/n)^2}{-\ell_n''(\theta_0)/n} + \ell_n''(\bar{\theta})/n \right) n(\theta_0 - \hat{\theta}_{ML})^2.$$

As argued above, $-\ell_n''(\bar{\theta})/n \xrightarrow{p} \mathcal{I}(\theta_0)$, and similarly, $-\ell_n''(\bar{\theta})/n \xrightarrow{p} \mathcal{I}(\theta_0)$. Since $-\ell_n''(\theta_0)/n \xrightarrow{p} \mathcal{I}(\theta_0)$ as well, expression in parentheses converges in probability to 0. Furthermore, $n(\theta_0 - \hat{\theta}_{ML})^2 \Rightarrow \chi^2(1)/\mathcal{I}(\theta_0)$ by the CMT. Thus, by Slutsky's theorem, $\zeta_{LM} - \zeta_{LR}$ converges in distribution (and hence also in probability) to zero. $\square$

Thus, we have shown that LR, Wald, and LM statistics are all asymptotically equivalent under the null hypothesis. However, they differ in finite samples.

Remark 2. An advantage of the LM statistic is that it only includes calculations based on the restricted estimator θ_0 . When is this particularly advantageous?

Example 1 (continued). Here $\mathcal{S}_n(\theta_0) = n(\bar{X}_n - \theta_0)/\sigma^2$, and $\mathcal{I}(\theta_0) = 1/\sigma^2$, so that $\zeta_{LM} = n(\bar{X}_n - \theta_0)^2/\sigma^2$, which is exactly equivalent to ζ_{LR} and ζ_W . $\boxtimes$

Example 2 (continued). We have

$$\mathcal{S}_n(\theta_0) = \sum_{i=1}^n X_i/\theta_0 - n = n(\bar{X}_n/\theta_0 - 1),$$

and since the Hessian is $-\ell''(\theta_0) = n\bar{X}_n/\theta_0^2$, $\zeta_{LM} = \zeta_W$. What about the alternative version of the score test? Since $\mathcal{I}(\theta_0) = 1/\theta_0$, we have

$$\tilde{\zeta}_{LM} = \frac{n^2(\bar{X}_n/\theta_0 - 1)^2}{n/\theta_0} = n\theta_0(\bar{X}_n/\theta_0 - 1)^2.$$

So the test rejects if $\tilde{\theta}_0(S/\tilde{\theta}_0 - 1)^2 \geq z_{1-\alpha/2}^2$. The acceptance region for S is therefore again an interval, $\tilde{\theta}_0 \pm z_{1-\alpha/2}\sqrt{\tilde{\theta}_0}$, and the exact finite-sample size can be computed analytically—it's plotted in Figure 1. The size is similar to that of the LR test, but without the spike at $\tilde{\theta}_0 = 1.921$ —instead, we have a spike close to zero. In the numerical example,

$$\zeta_{LM} = 100 \cdot 6 \cdot (5/6 - 1)^2 = 100/6 \approx 16.67,$$

which gives the biggest p -value among the three tests. $\boxtimes$

4. GENERALIZATIONS AND SUMMARY

The three members of the trinity differ in how they expand the log-likelihood function:

1. The LR test statistic is given by twice the actual difference between $\ell_n(\hat{\theta}_{ML})$ and $\ell_n(\theta_0)$, $\zeta_{LR} = 2(\ell_n(\hat{\theta}_{ML}) - \ell_n(\theta_0))$.
2. The Wald test uses a quadratic approximation to the log-likelihood at $\hat{\theta}_{ML}$, and compares $\ell_n(\hat{\theta}_{ML})$ to its value at θ_0 , as predicted by the approximation. Since $\ell'_n(\hat{\theta}_{ML}) = 0$, the approximation predicts that at θ_0 , the log-likelihood value will be $\hat{\ell}_W(\theta_0) = \ell_n(\hat{\theta}_{ML}) + \frac{1}{2}(\hat{\theta}_{ML} - \theta_0)^2 \ell''_n(\hat{\theta}_{ML})$. Consequently, the Wald statistic is $\zeta_W = 2(\ell_n(\hat{\theta}_{ML}) - \hat{\ell}_W(\theta_0)) = (-\ell''_n(\hat{\theta}_{ML}))(\hat{\theta}_{ML} - \theta_0)^2$. Intuitively, if $\hat{\theta}_{ML}$ is far from θ_0 , relative to the curvature of the log-likelihood at $\hat{\theta}_{ML}$ (as measured by the second derivative ℓ''_n), then we expect the objective function to be a lot lower at θ_0 , so we reject.
3. The LM test uses a quadratic approximation to the log-likelihood at θ_0 , and compares $\ell_n(\theta_0)$ to the maximum value of the log-likelihood, as predicted by the expansion. Expanding the log-likelihood at θ_0 gives the approximation

$$\hat{\ell}_{LM}(\theta) = \ell_n(\theta_0) + (\theta - \theta_0)S_n(\theta_0) + \frac{1}{2}(\theta - \theta_0)^2 \ell''_n(\theta_0).$$

The first-order condition is $S_n(\theta_0) + (\theta - \theta_0)\ell''_n(\theta_0) = 0$. Thus, $\hat{\ell}_{LM}(\theta)$ is maximized at $\theta_{\max} = -(\ell''_n(\theta_0))^{-1}S_n(\theta_0) + \theta_0$, with the objective given by $\hat{\ell}_{LM}(\theta_{\max}) = \ell_n(\theta_0) + \frac{1}{2}(-\ell''_n(\theta_0))^{-1}S_n(\theta_0)^2$. Hence,

$$\zeta_{LM} = 2(\hat{\ell}_{LM}(\theta_{\max}) - \ell_n(\theta_0)) = -\ell''_n(\theta_0)^{-1}S_n(\theta_0)^2.$$

Intuitively, if $S_n(\theta_0)$, the slope at θ_0 , is large relative to the curvature of the log-likelihood at θ_0 , then we expect that $\ell_n(\theta_0)$ is way below the peak of the log-likelihood, so we reject.

Figure 2 depicts quadratic approximations used by the trinity graphically.

If the log-likelihood is exactly quadratic—as in Example 1—then there is no approximation error in the Wald and LR expansions, and $\zeta_W = \zeta_{LR} = \zeta_{LM}$.

Consider now the general case in which the density $f(x | \theta)$ is indexed by a vector $\theta \in \Theta \subseteq \mathbb{R}^d$, and we are interested in testing the null $H_0: g(\theta) = 0$ against the alternative $H_1: g(\theta) \neq 0$. More precisely, $H_0: \theta \in \Theta_0 = \{\theta \in \Theta: g(\theta) = 0\}$, while $H_1: \theta \in \Theta \setminus \Theta_0$. The function $g: \mathbb{R}^d \rightarrow \mathbb{R}^p$ collects smooth, possibly nonlinear restrictions. These restrictions are assumed to be non-redundant in the sense that we cannot drop any one g_j without changing Θ_0 . This means that the Jacobian G with elements $G_{ij} = \partial g_i(\theta) / \partial \theta_j$ has full row rank.

Let $\hat{\theta}_0 = \operatorname{argmax}_{\theta \in \Theta_0} \mathcal{L}_n(\theta | X)$ denote the MLE in the restricted model, and let $\hat{\theta}_{ML} = \operatorname{argmax}_{\theta \in \Theta} \mathcal{L}_n(\theta | X)$ denote the unrestricted estimate.

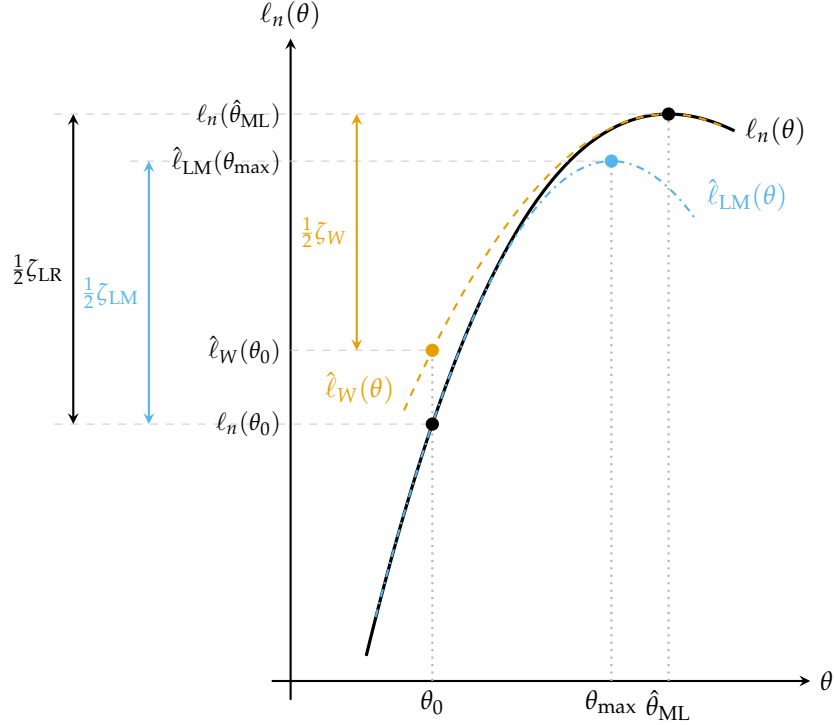


Figure 2: Geometric comparison of the LR, Wald, and LM tests.

As in the scalar case, we either compare the values of the log-likelihood at $\hat{\theta}_{ML}$ to its value at $\hat{\theta}_0$ directly, or use quadratic expansions, which yields the general form of the trinity of tests:

$$\begin{aligned}\zeta_{LR} &= 2(\ell_n(\hat{\theta}_{ML}) - \ell_n(\hat{\theta}_0)) \Rightarrow \chi^2(p) \\ \tilde{\zeta}_W &= (\hat{\theta}_{ML} - \hat{\theta}_0)' \left(-\nabla^2 \ell_n(\hat{\theta}_{ML}) \right) (\hat{\theta}_{ML} - \hat{\theta}_0) \Rightarrow \chi^2(p) \\ \zeta_{LM} &= \mathcal{S}_n(\hat{\theta}_0)' (-\nabla^2 \ell_n(\hat{\theta}_0))^{-1} \mathcal{S}_n(\hat{\theta}_0) \Rightarrow \chi^2(p),\end{aligned}$$

and all of them are asymptotically equivalent to each other. For the score and the Wald tests, one can use the information matrix instead of the Hessian: any one of $-\nabla^2 \ell_n(\hat{\theta}_{ML})$, $-\nabla^2 \ell_n(\hat{\theta}_0)$, $n\mathcal{I}(\hat{\theta}_0)$, or $n\mathcal{I}(\hat{\theta}_{ML})$ will do, depending on what's most convenient. One can also use an estimator based on the outer product of the scores, $\sum_{i=1}^n \nabla \log f(X_i | \theta) \cdot (\nabla \log f(X_i | \theta))'$, evaluated at $\hat{\theta}_{ML}$ or $\hat{\theta}_0$.

Why did I write the Wald test with a tilde? One advantage of the scalar version of the Wald test was that it didn't involve any calculations under the null. The vector version $\tilde{\zeta}_W$ loses this advantage, because it requires computation of $\hat{\theta}_0$. However, the Wald test does have another form that retains this advantage. In particular, by the delta method, $\sqrt{n}(g(\hat{\theta}_{ML}) - g(\theta)) \Rightarrow \mathcal{N}(0, GI(\theta)^{-1}G')$. We can estimate the Jacobian by

$\hat{G}_{ij} = \partial g_i(\hat{\theta}_{ML}) / \partial \theta_j$. Since $g(\theta) = 0$ under the null, we have

$$\zeta_W = \sqrt{n}g(\hat{\theta}_{ML})' \left(\hat{G}\mathcal{I}(\hat{\theta}_{ML})^{-1}\hat{G}' \right)^{-1} \sqrt{n}g(\hat{\theta}_{ML}) \Rightarrow \chi^2(p).$$

This version seems like a better default, so we write it with no tilde.

A leading special case of ζ_W is when the restrictions are linear, $g(\theta) = R\theta - r_0$. In this case, the Wald test is particularly easy,

$$\zeta_W = \sqrt{n}(R\hat{\theta}_{ML} - r_0)' \left(R\mathcal{I}(\hat{\theta}_{ML})^{-1}R' \right)^{-1} \sqrt{n}(R\hat{\theta}_{ML} - r_0).$$

A leading special case of this special case obtains when $\theta = (\beta, \eta)$, η is a nuisance parameter of dimension $d - p$, $\beta \in \mathbb{R}^p$, and we're interested in a two-sided test of the hypothesis $\beta = \beta_0$. Then, under the null,

$$\zeta_W = \sqrt{n}(\hat{\beta}_{ML} - \beta_0)' \hat{V}_{\beta\beta}^{-1} \sqrt{n}(\hat{\beta}_{ML} - \beta_0) \Rightarrow \chi^2(p),$$

where $\hat{V}_{\beta\beta}$ is the upper $p \times p$ block of the inverse information matrix $\mathcal{I}(\hat{\theta}_{ML})^{-1}$. If θ is the true parameter value, this upper block converges in probability to the (β, β) block of $\mathcal{I}(\theta)^{-1}$, given by

$$[\mathcal{I}(\theta)^{-1}]_{\beta\beta} = \left(\mathcal{I}_{\beta\beta}(\theta) - \mathcal{I}_{\beta\eta}(\theta)\mathcal{I}_{\eta\eta}(\theta)^{-1}\mathcal{I}_{\eta\beta}(\theta) \right)^{-1}.$$

The matrix $\hat{V}_{\beta\beta}$ estimates the asymptotic variance of $\sqrt{n}(\hat{\beta}_{ML} - \beta)$, since $\sqrt{n}(\hat{\theta}_{ML} - \theta) \Rightarrow \mathcal{N}(0, \mathcal{I}(\theta)^{-1})$. Note that in general, $\hat{V}_{\beta\beta}^{-1}$ is *not* equal to $\mathcal{I}_{\beta\beta}(\hat{\theta}_{ML})$, unless the information matrix is block-diagonal, so that $\mathcal{I}_{\beta\eta}(\hat{\theta}_{ML}) = 0$.

Remark 3. One can also show that the power against any fixed alternative $\theta_1 \in \Theta$ for any of these tests converges to one as $n \rightarrow \infty$. When the power function satisfies $\pi(\theta_1) \rightarrow 1$ as $n \rightarrow \infty$, we say that the test is *consistent* against the alternative $H_1: \theta = \theta_1$.

Remark 4. An attractive feature of the Wald test ζ_W is that it works whenever we have an asymptotically normal estimator $\hat{\theta}$ of θ with consistently estimable asymptotic variance. The model doesn't even have to be fully parametric.

5. OPTIMALITY

We have derived three tests for restrictions in parametric models that are asymptotically equivalent under the null. But what about optimality? We motivated the Wald statistic by observing that $\hat{\theta}_{ML} \overset{\text{asympt}}{\sim} \mathcal{N}(\theta, \mathcal{I}(\hat{\theta}_{ML})^{-1}/n)$. Since $\hat{\theta}_{ML}$ is asymptotically efficient, this suggests that when $\theta = (\beta, \eta)$, with β scalar, the problem of testing the null $H_0: \beta = \beta_0$ against the alternative $H_1: \beta \neq \beta_0$ should in large samples be equivalent to the problem of testing a normal mean. We saw in previous lectures that the UMP unbiased test for this problem is the two-sided z-test. This suggests that the two-sided Wald test

should be asymptotically UMP unbiased. This is indeed true, and what's more, the LR and LM variants also share this optimality property.

Just like the large-sample optimality of MLE as a point estimator, properly stating this large-sample optimality of the trinity of tests requires quite a bit of machinery, best left for a second-year course. But loosely speaking, the optimality result means that for values of θ over a shrinking neighborhood of θ_0 , (i) the power functions of all three tests are the same, and (ii) no other test that is asymptotically unbiased and controls size in large samples has a power function that's better. This is the real justification for using these tests.

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